

Aging of hepatitis C virus (HCV)-infected persons in the United States: a multiple cohort model of HCV prevalence and disease progression.

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BACKGROUND & AIMS: The prevalence of chronic hepatitis C (CH-C) remains high and the complications of infection are common. Our goal was to project the future prevalence of CH-C and its complications. METHODS: We developed a multicohort natural history model to overcome limitations of previous models for predicting disease outcomes and benefits of therapy. RESULTS: Prevalence of CH-C peaked in 2001 at 3.6 million. Fibrosis progression was inversely related to age at infection, so cirrhosis and its complications were most common after the age of 60 years, regardless of when infection occurred. The proportion of CH-C with cirrhosis is projected to reach 25% in 2010 and 45% in 2030, although the total number with cirrhosis will peak at 1.0 million (30.5% higher than the current level) in 2020 and then decline. Hepatic decompensation and liver cancer will continue to increase for another 10 to 13 years. Treatment of all infected patients in 2010 could reduce risk of cirrhosis, decompensation, cancer, and liver-related deaths by 16%, 42%, 31%, and 36% by 2020, given current response rates to antiviral therapy. CONCLUSIONS: Prevalence of hepatitis C cirrhosis and its complications will continue to increase through the next decade and will mostly affect those older than 60 years of age. Current treatment patterns will have little effect on these complications, but wider application of antiviral treatment and better responses with new agents could significantly reduce the impact of this disease in coming years.